

POPULATION CODING IN ZEBRAFISH OPTIC TECTUM



POPULATION CODING IN ZEBRAFISH OPTIC TECTUM

Introduction: Recording Zebrafish optic tectum activity

Decoding: Population vector, Template Matching, Bayesian ML

Encoding: Predictive coding

Variability: How informative Signal and Noise Correlations could be ?

POPULATION CODING IN ZEBRAFISH OPTIC TECTUM

Introduction: Recording Zebrafish optic tectum activity

Zebrafish

Small vertebrate, ideal scale to study precise network patterns.
Gcamp fishes are enabled for calcium imaging neural activity acquisition.

Optic Tectum : A RETINOTOPICAL ORGANISATION

Optic terminals are distributed in an orderly fashion over tectum so as to make a map of the retina. The retinotectal map is organized so that the front, back, top and bottom of the visual field seen by one eye is represented at the front, back, top and bottom of the opposite tectal lobe.

RF size : 25-39 degrees

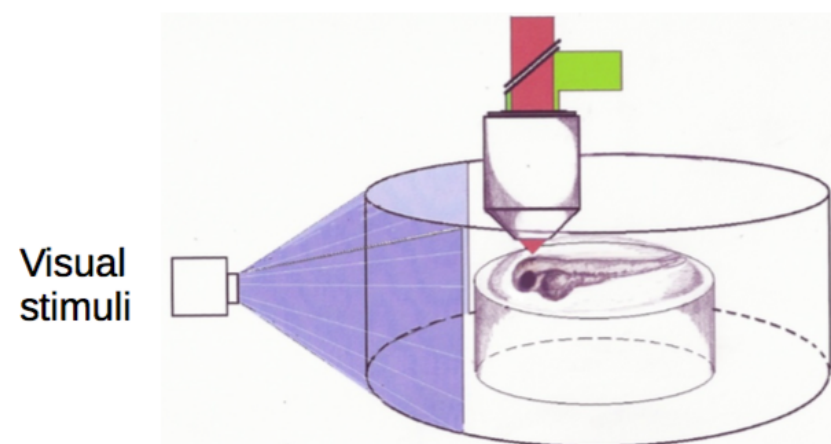
POPULATION CODING IN ZEBRAFISH OPTIC TECTUM

Introduction: Recording Zebrafish optic tectum activity

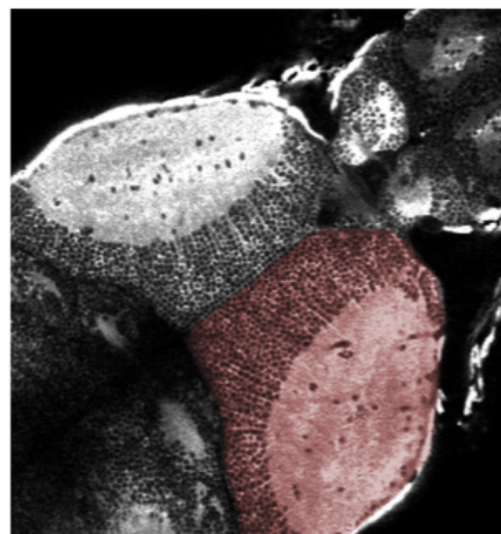
Arturo Torres-Herraez, Jonathan Boulanger-Weill, Adrien Jouary and German Sumbre

Institut de Biologie de l'Ecole Normale Supérieure, INSERM, CNRS

Two-photon microscope



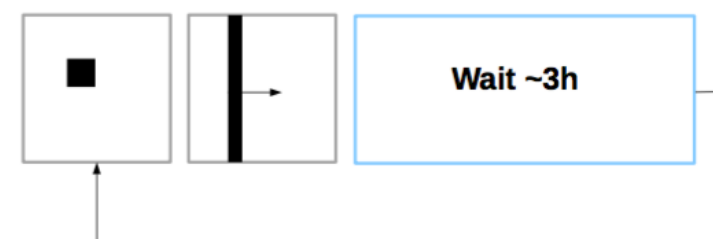
Imaging of HuC:GCaMP5G 7days post fertilization larva



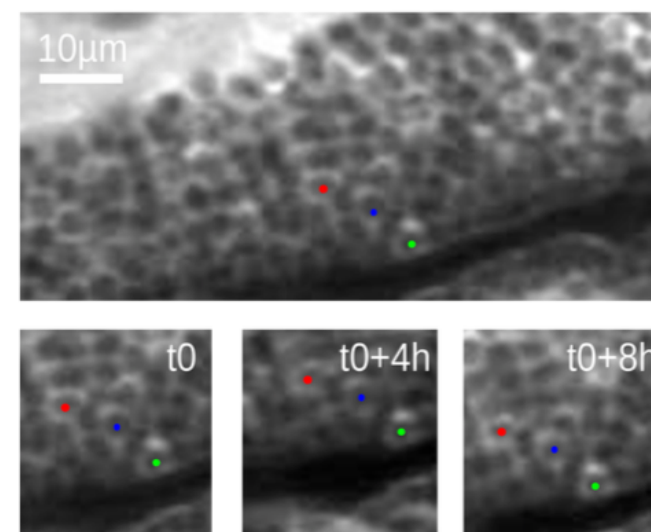
Experimental protocol

Stimulation ~1h :

- Bars in each 4 cardinal direction,
4 repeats of each
- 9*4 squares of 5° length, separated by 5° blanks
20 random repeats of each



Manual neuronal recognition



POPULATION CODING IN ZEBRAFISH OPTIC TECTUM

Introduction: Recording Zebrafish optic tectum activity

Direction Selectivity only.

Stimuli coded by angle of provenance of the moving bar:

Up $\pi/2$

Right 0

Down $-\pi/2$

Left $-\pi$

POPULATION CODING IN ZEBRAFISH OPTIC TECTUM

Decoding: Population Vector Method

Neuron's preferred direction is defined as the angle of following vector .

$$R_n = \frac{\sum_{k=0..K} r_k^n \cdot e^{i\theta_k}}{\sum_{k=0..K} r_k^n}$$

where $R_n(\theta)$ is the mean of neuron n responses to stimulus θ . K = 4 (4 different stimuli)

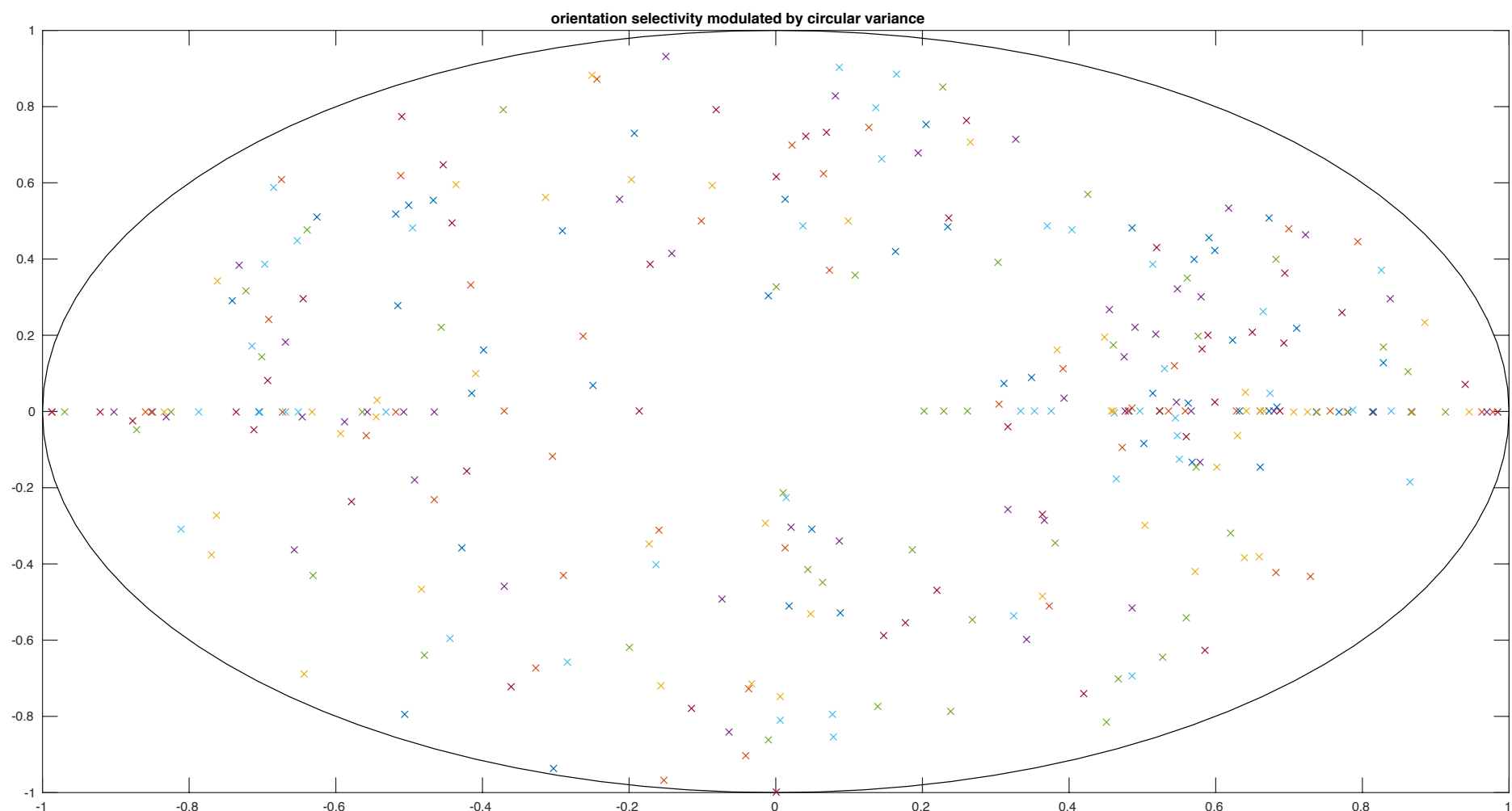
Circular variance :

$$V(n) = 1 - ||R_n||$$

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Decoding: Population Vector Method

For each trial, the activity of each neuron is represented as a vector. The angle (θ) represents the neuron's preferred direction and the magnitude (r) corresponds to its activity level (dF/F_0).

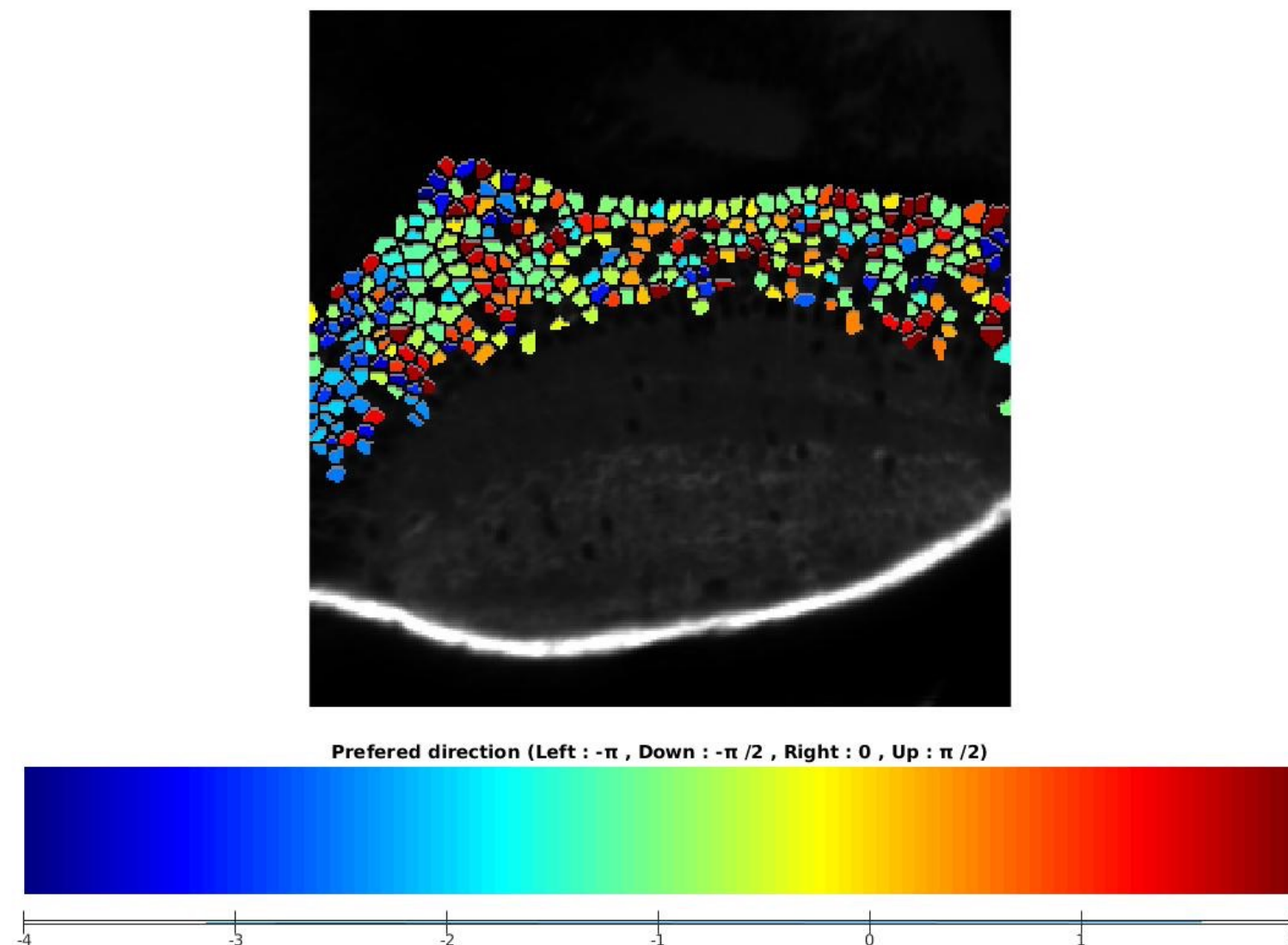


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Decoding: Population Vector Method

For decoding, as we deal with only four discrete angle values, one can choose as the preferred direction the angle that maximises the across-trial mean response computed for each one of the stimuli.

For each θ , $R_n(\theta)$ is the mean response of neuron n for a θ oriented bar stimulus.

Preferred direction is defined for decoding as :

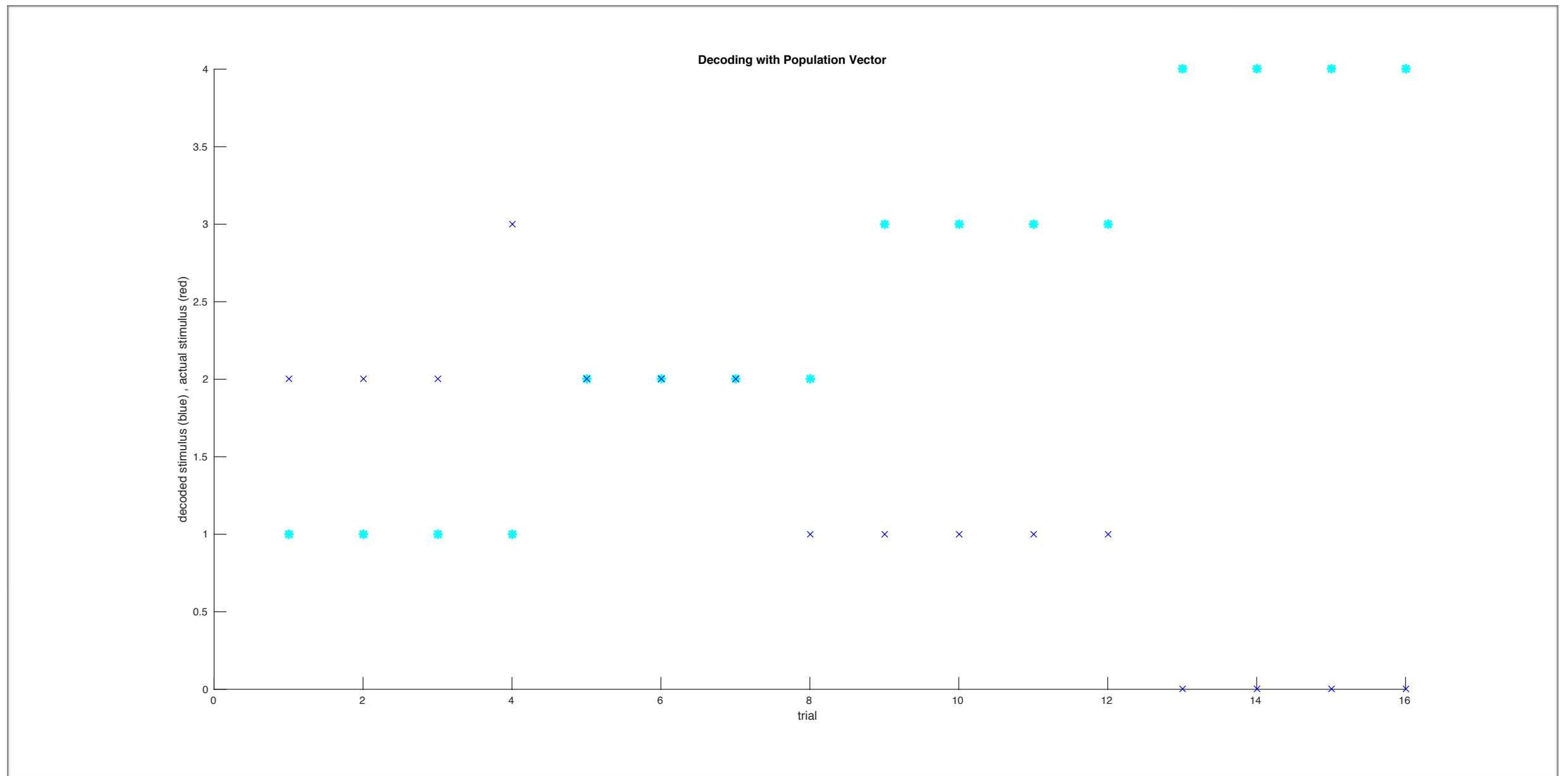
$$D_n = \operatorname{argmax}_{\theta}(\max(R_n(\theta)))$$

Decoded direction :

$$DD(t) = \operatorname{angle}\left(\frac{\sum_n r_n(t) \cdot e^{iD_n}}{\sum_n r_n(t)}\right)$$

POPULATION CODING IN ZEBRAFISH OPTIC TECTUM

Decoding: Population Vector Method



POPULATION CODING IN ZEBRAFISH OPTIC TECTUM

Decoding: Template Matching Method

For each presented stimulus angle, a response template (RS) corresponding to the over trials average neural activity is computed neuron per neuron.

It is given by :

$$RT(\theta, n) = \frac{\sum_{k=1}^{N_{trial}} r_n(\theta, k)}{N_{trial}}$$

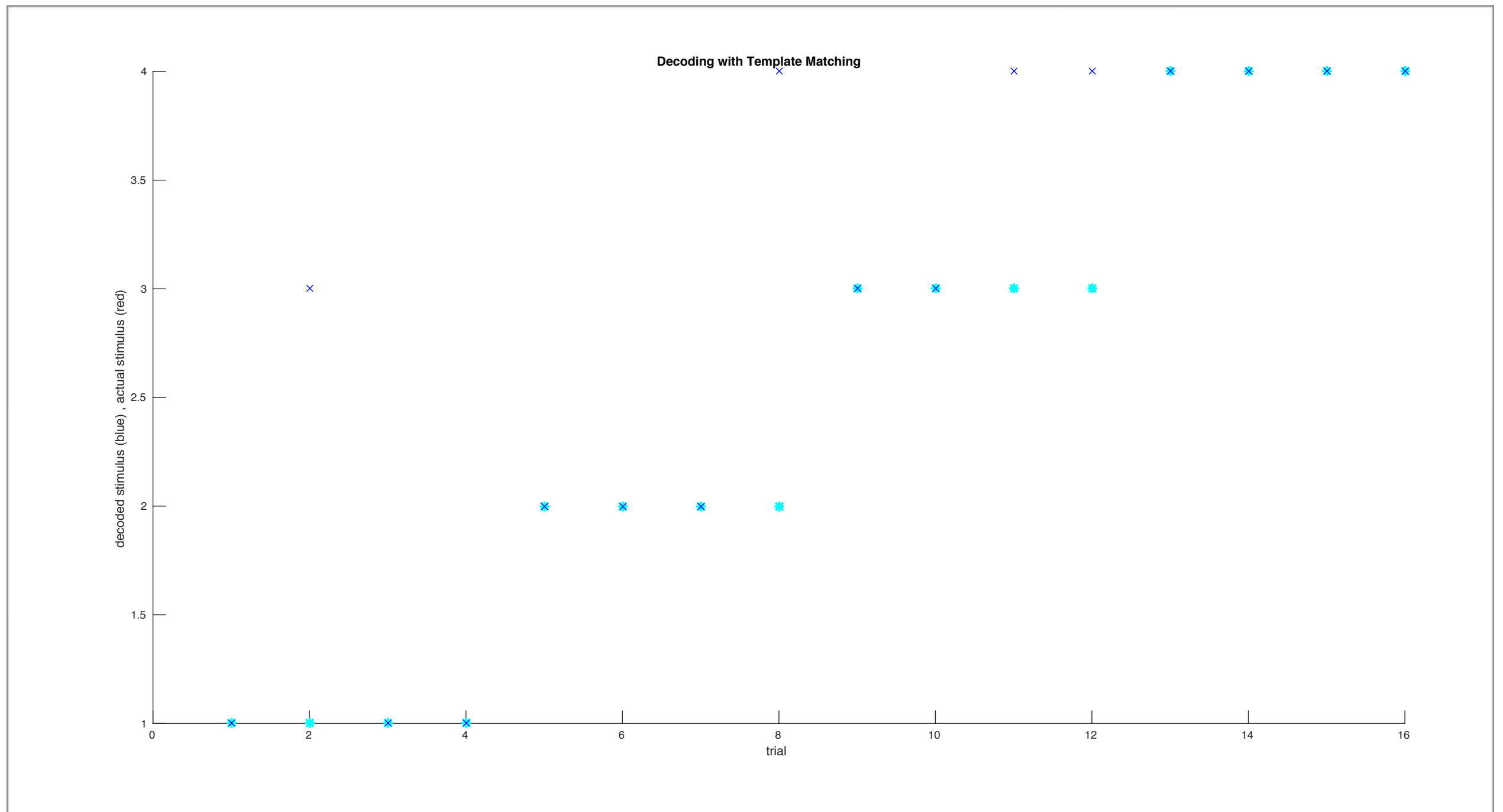
The similarity of the total vector $RT(\theta)$ for a given angle to the actual population activity is given by a similarity index, expressed as :

$$\Theta_{\theta} = \frac{\sum_n R_n^{stim} \cdot RT(\theta, n)}{||R^{stim}|| \cdot ||RT(\theta)||}$$

Decoded direction is defined as angle θ with maximum similarity index

POPULATION CODING IN ZEBRAFISH OPTIC TECTUM

Decoding: Template Matching Method



POPULATION CODING IN ZEBRAFISH OPTIC TECTUM

Decoding: Normalized Template Matching Method

Same method can be implemented with z-scored dF/F rather than the raw activity.

For any time point, and for each neuron, its activity is re-written as :

$$Z_n(t) = \frac{R_n(t) - E_t(R_n(t))}{\sigma(R_n)}$$

Then, following the same framework, $ZT(\theta)$ can be computed.

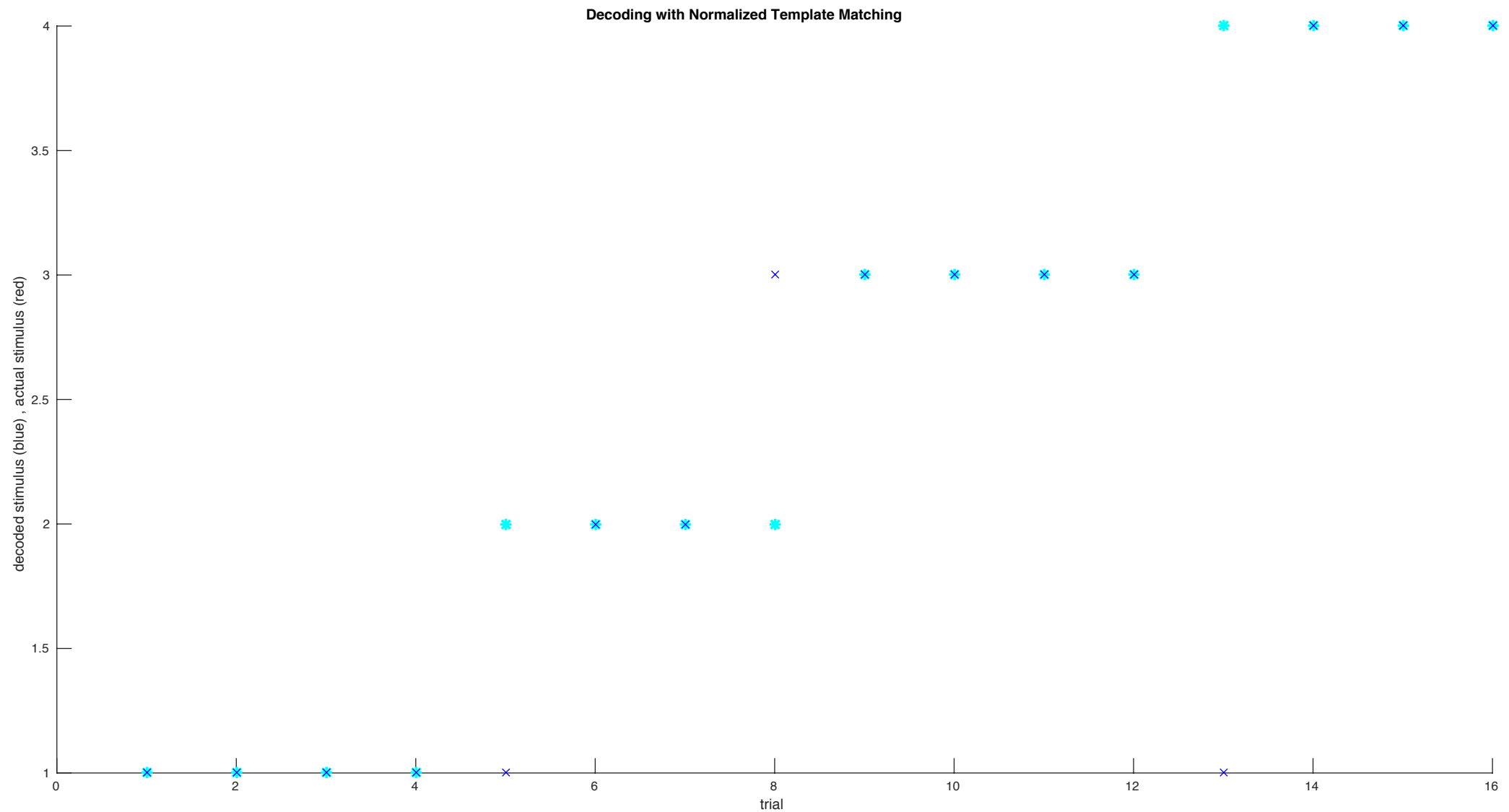
In this context, similarity index is given by :

$$\Theta_\theta = \frac{\sum_n Z_n^{stim} \cdot ZT(\theta, n)}{||Z^{stim}|| \cdot ||ZT(\theta)||}$$

Decoded direction is defined as angle θ with maximum similarity index

POPULATION CODING IN ZEBRAFISH OPTIC TECTUM

Decoding: Normalized Template Matching Method



POPULATION CODING IN ZEBRAFISH OPTIC TECTUM

Decoding: Bayesian Maximum-Likelihood

$$P(\theta|R_{pop}) = \frac{P(R_{pop}|\theta).P(\theta)}{P(R_{pop})}$$

Uniform prior.

$P(R_{pop})$ equal for all stimulus direction (no angle dependance).

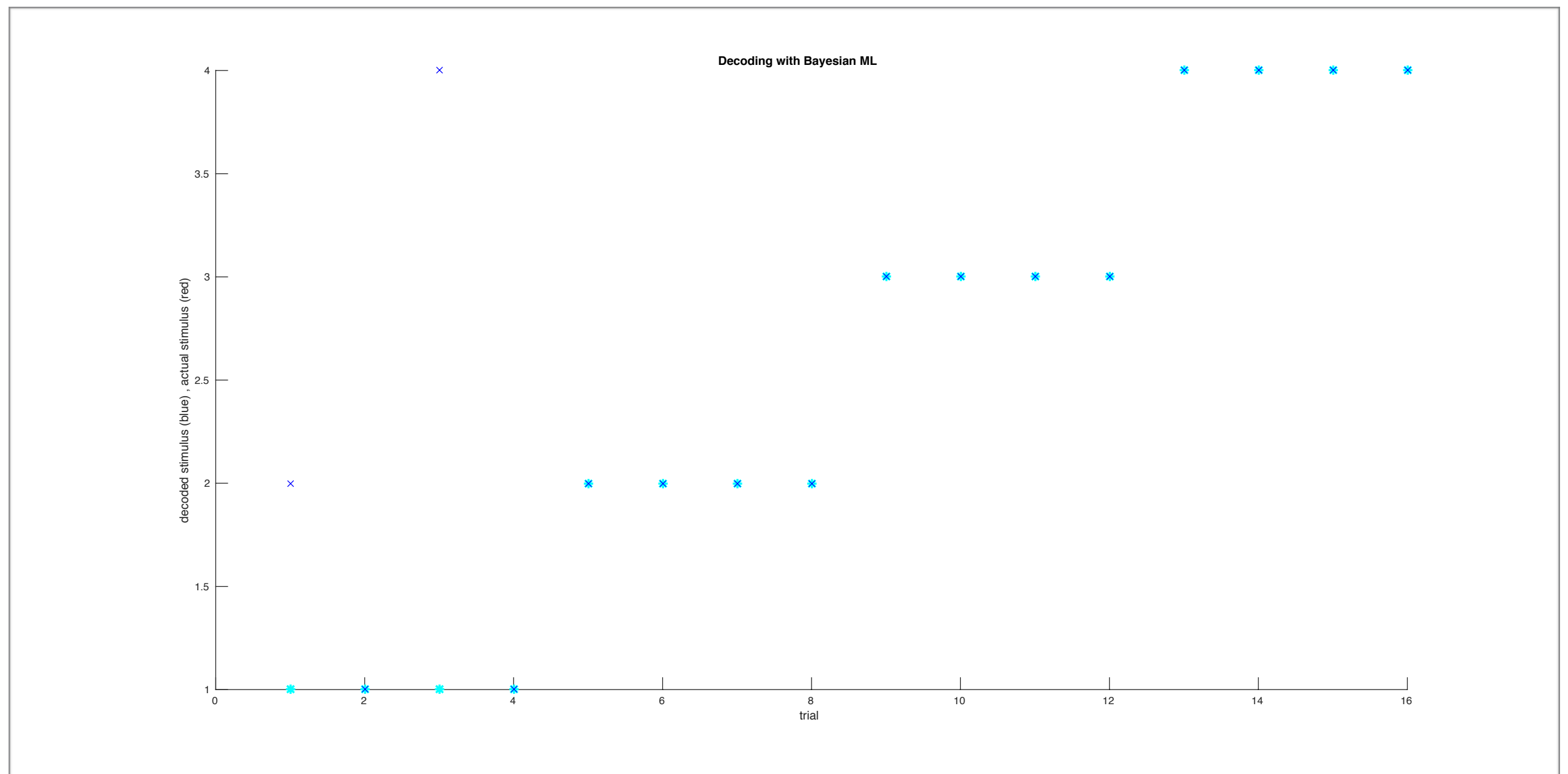
For a given angle, the posterior probability is proportional to the likelihood.

The likelihood, for a given angle and a given neuron, proportional to $P(\theta|R_n)$ via Bayle's rule.

$$P(\theta|R_{pop}) \propto \prod_{n=1}^N P(\theta|R_n)$$

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Decoding: Bayesian Maximum-Likelihood



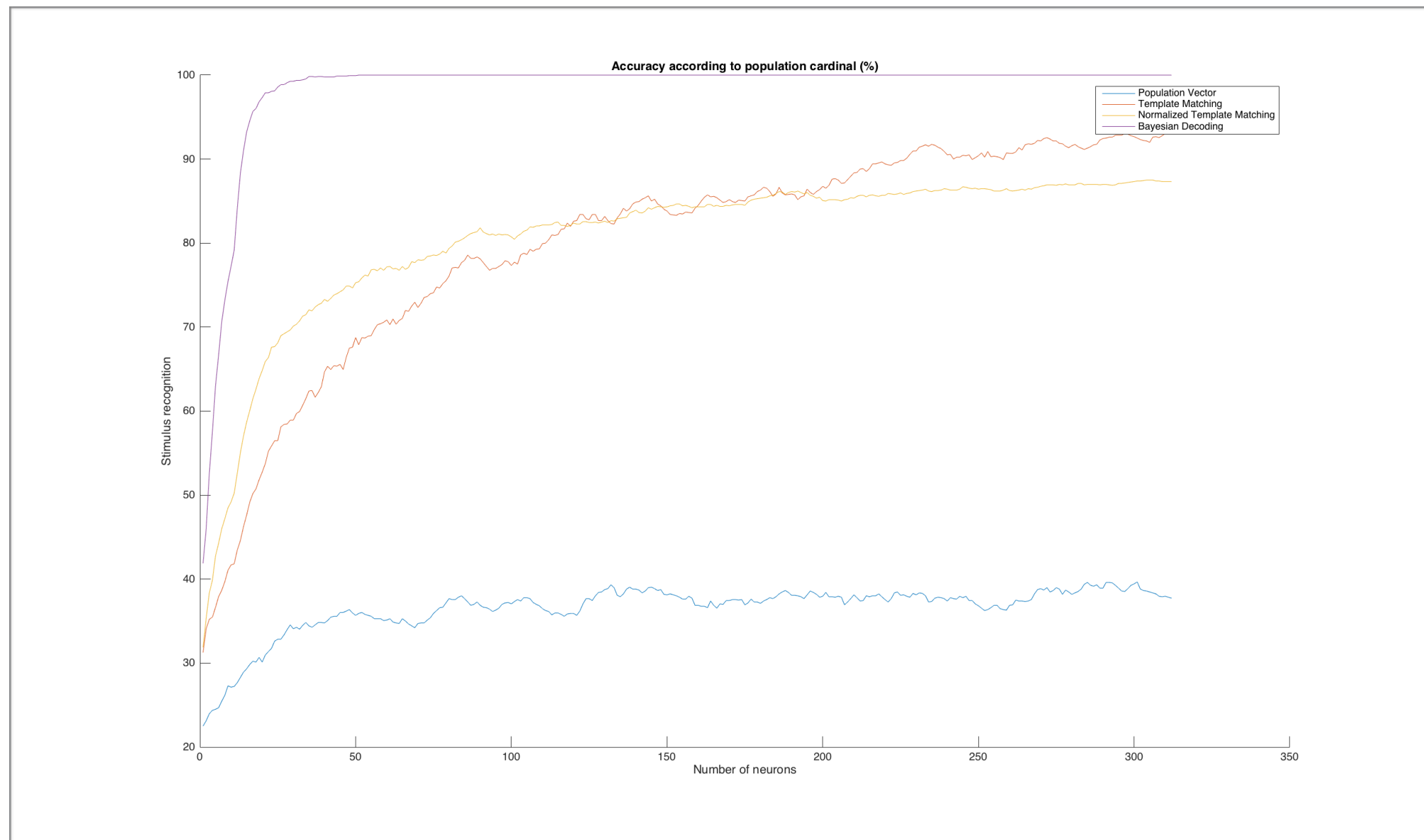
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Decoding: Comparing performances

I compared accuracy of those different algorithms, and looked at how it evolved as the number of neurons integrated in the model changed.

POPULATION CODING IN ZEBRAFISH OPTIC TECTUM

Decoding: Comparing performances



PREDICTIVE CODING : SPIKING NETWORK

Model presented by Boerlin, Machens, Deneve et al, 2013.

$$\dot{\mathbf{x}} = \mathbf{A} \mathbf{x} + \mathbf{c}(t).$$

$$\dot{\hat{\mathbf{x}}} = -\lambda_d \hat{\mathbf{x}} + \mathbf{\Gamma} \mathbf{o}(t),$$

$$\dot{\mathbf{r}} = -\lambda_d \mathbf{r} + \lambda_d \mathbf{o}(t).$$

$$E(t) = \int_0^t du (\|\mathbf{x}(u) - \hat{\mathbf{x}}(u)\|_2^2 + v \|\mathbf{r}(u)\|_1 + \mu \|\mathbf{r}(u)\|_2^2),$$

the firing rule

$$V_i(t) > T_i$$

with

$$V_i(t) = \mathbf{\Gamma}_i^T (\mathbf{x}(t) - \hat{\mathbf{x}}(t)) - \mu \lambda_d r_i(t)$$

$$T_i = \frac{v \lambda_d + \mu \lambda_d^2 + \|\mathbf{\Gamma}_i\|^2}{2}$$

Model tested for same number of neurons than recorded population cardinal.
Preferred direction used for population vector decoding where chosen as kernels
orientations.

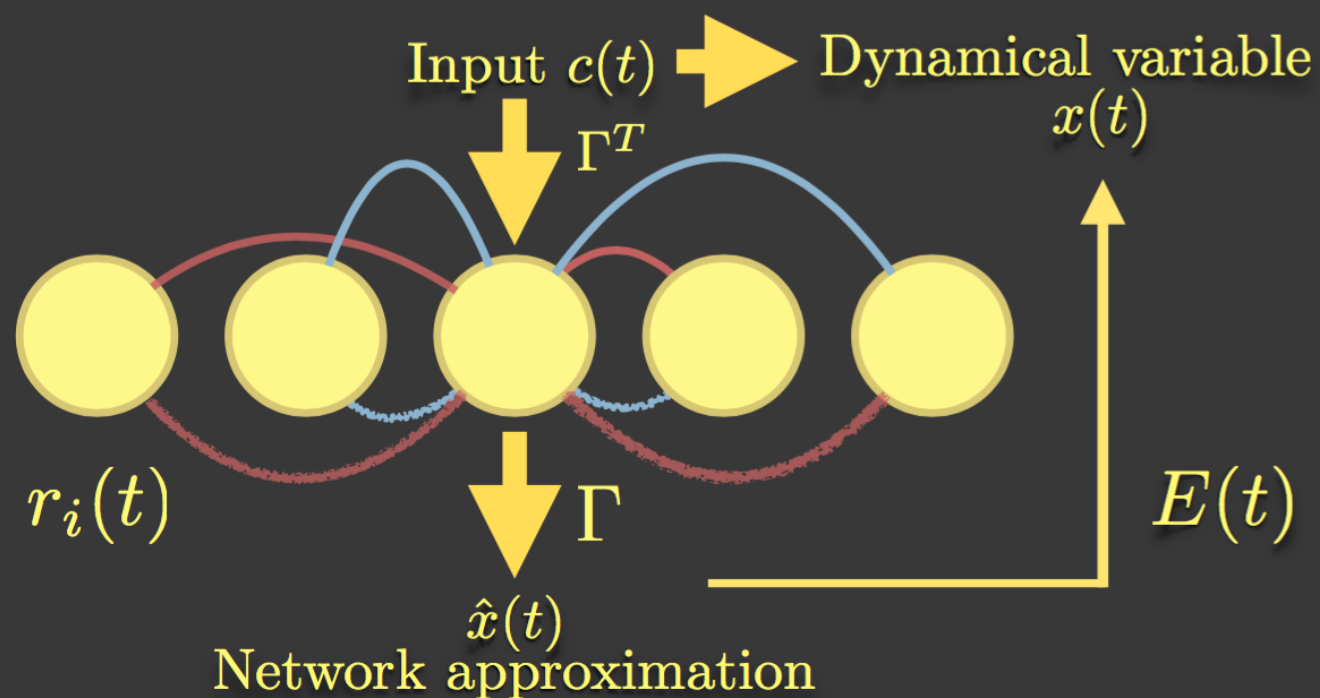
Balance between negative and positive weights in kernel for each direction.

Stimulus : encoded by a binary 4-dimension vector.

PREDICTIVE CODING : SPIKING NETWORK

Model presented by Boerlin, Machens, Deneve et al, 2013.

$$E(t) = \int_0^t (||x(u) - \hat{x}(u)||_2^2 + \nu ||r(u)||_1 + \mu ||r(u)||_2^2)$$



$$V_i(t) > T_i$$

- $o_i(t) = \delta(t)$
- $\dot{\hat{x}} = -\lambda_d \hat{x} + \Gamma o(t)$
- $\dot{r} = -\lambda_d r + \lambda d o(t)$
- $E(t)$ updated

x dynamics:

$$-\dot{x} = -\lambda_s x + c(t)$$

PREDICTIVE CODING : SPIKING NETWORK

Model presented by Boerlin, Machens, Deneve et al, 2013.

Dynamic variable implements a stimulus tracking.

$$J = 4$$

$$A = -0.9 \times I_J$$

$$\lambda_d = 0.61$$

$$\mu = 10$$

$$\nu = 1$$

PREDICTIVE CODING : SPIKING NETWORK

Model presented by Boerlin, Machens, Deneve et al, 2013.

Stimulus : encoded by a binary 4-dimension vector.

$$c(t) \in \{0, 1\}^4$$

$$\begin{array}{c} \textit{Up} \\ c = [1, 0, 0, 0] \end{array}$$

$$\begin{array}{c} \textit{Left} \\ c = [0, 0, 0, 1] \end{array}$$

$$\begin{array}{c} \textit{Right} \\ c = [0, 1, 0, 0] \end{array}$$

$$\begin{array}{c} \textit{Down} \\ c = [0, 0, 1, 0] \end{array}$$

PREDICTIVE CODING : SPIKING NETWORK

Model presented by Boerlin, Machens, Deneve et al, 2013.

Kernel choice

Find W such that

$$r = W.x$$

where:

x is the input's descriptive variable (position, angle)

$$r \in \mathbb{R}^{N_{neurons} \times T}$$

$$\text{State } \Gamma_i = f(W_i \in \mathbb{R}^{N_{features}(x)})$$

Underlying idea : when error projection is big enough on one neurone kernel, the neurone spikes.

So Γ can be seen as a set of vectors of preferred directions for network's correction game.

Each neurone has its preferred direction.

PREDICTIVE CODING : SPIKING NETWORK

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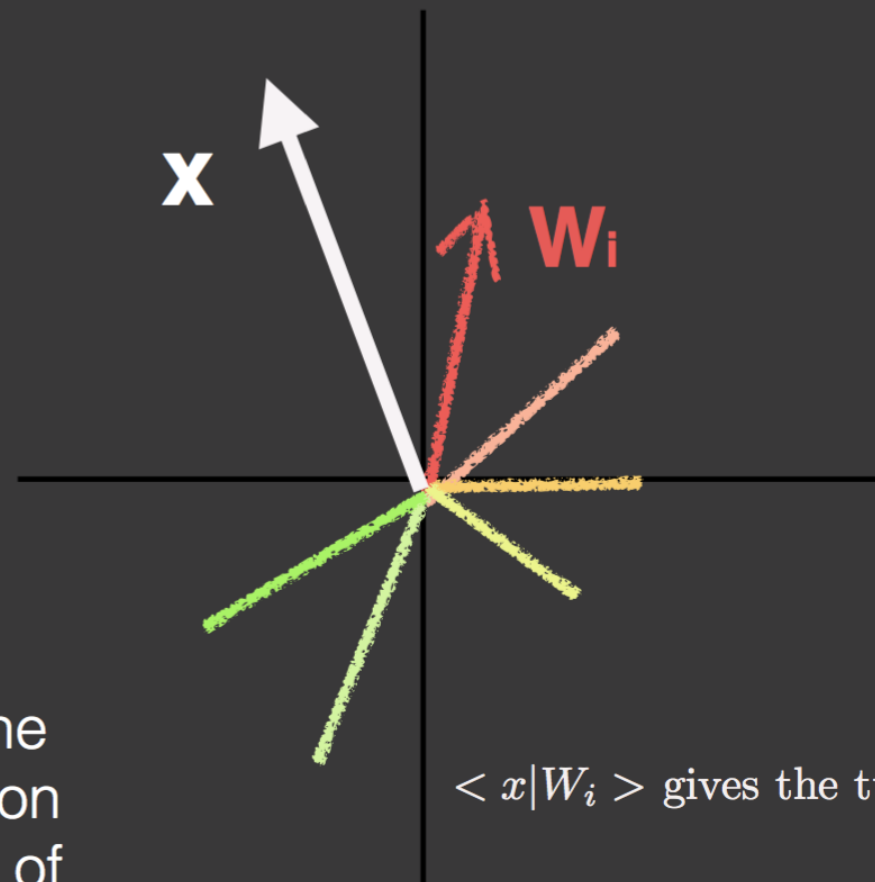
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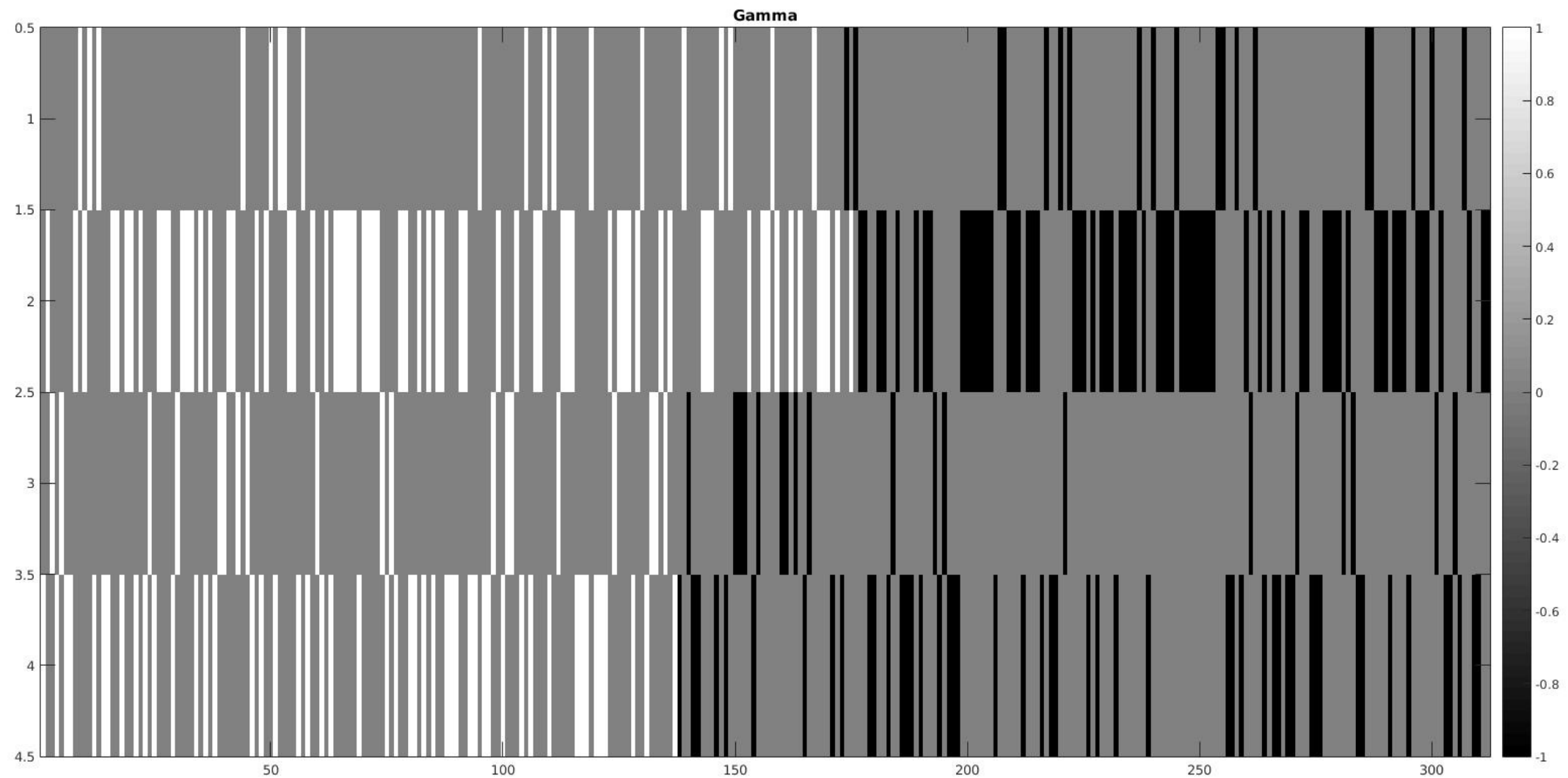
Tuning curve for neurone i should be given by the scalar product of its theoretical preferred direction (according to the network) with the whole range of stimulus.



PREDICTIVE CODING : SPIKING NETWORK

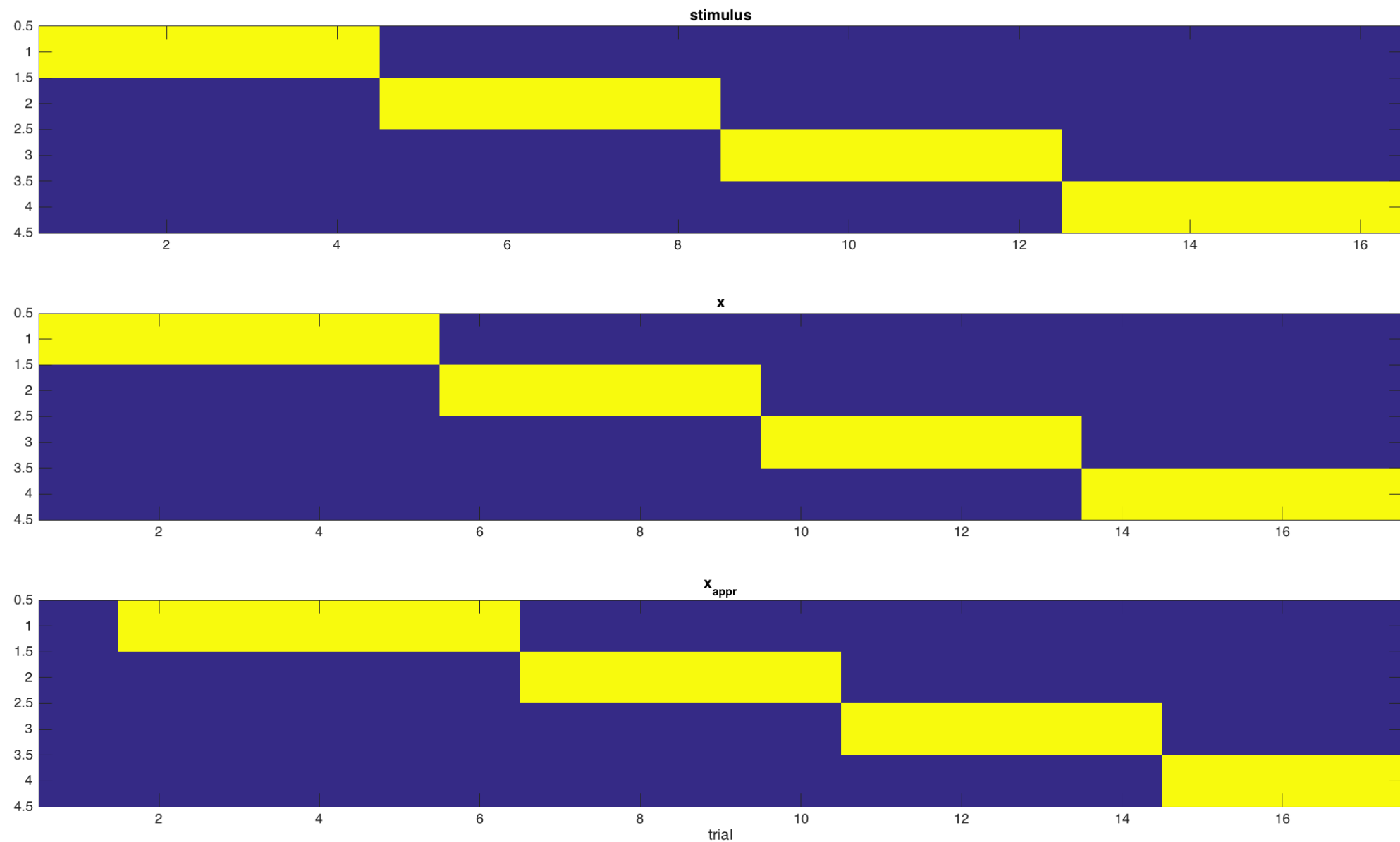
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Kernel choice



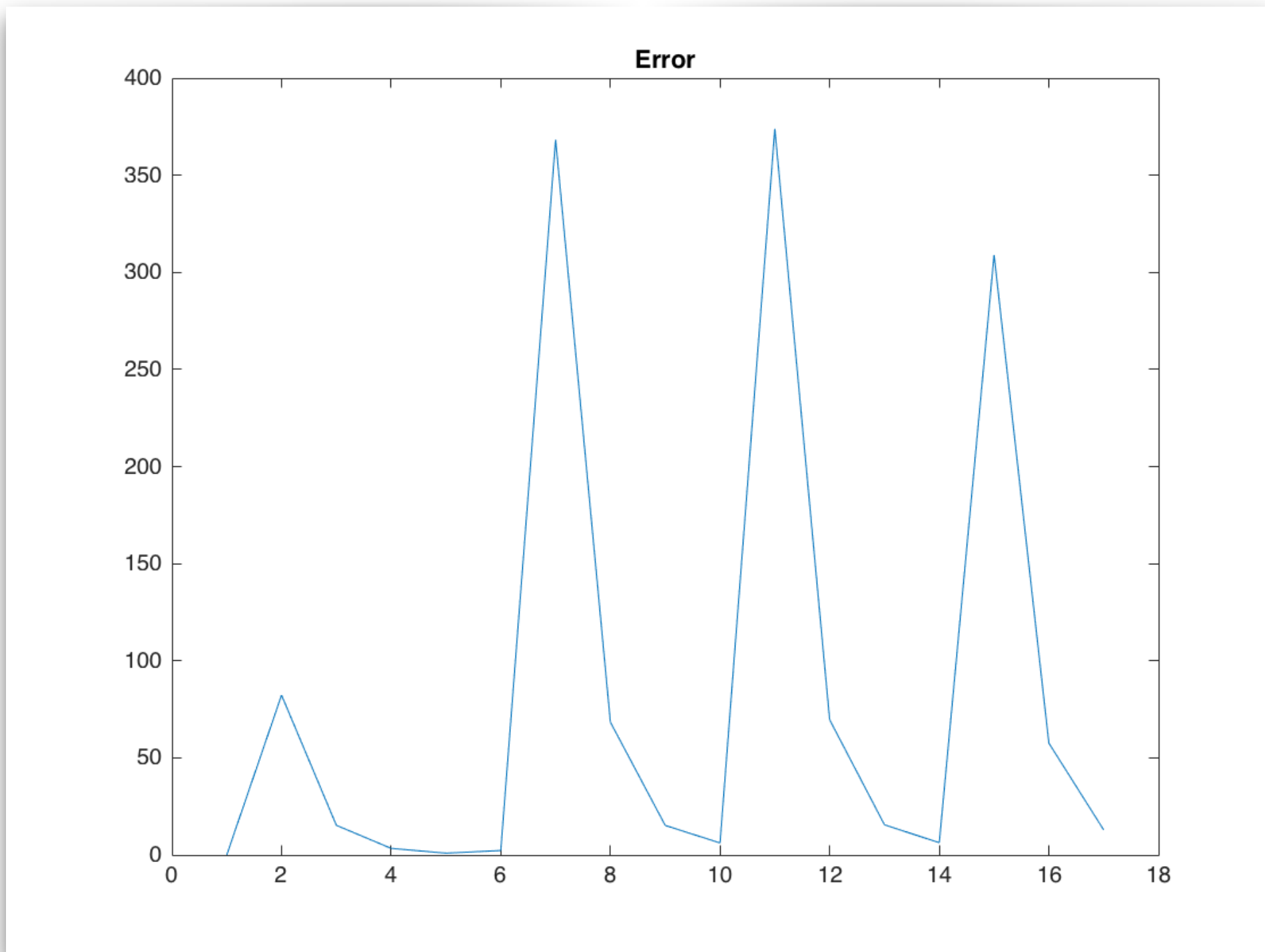
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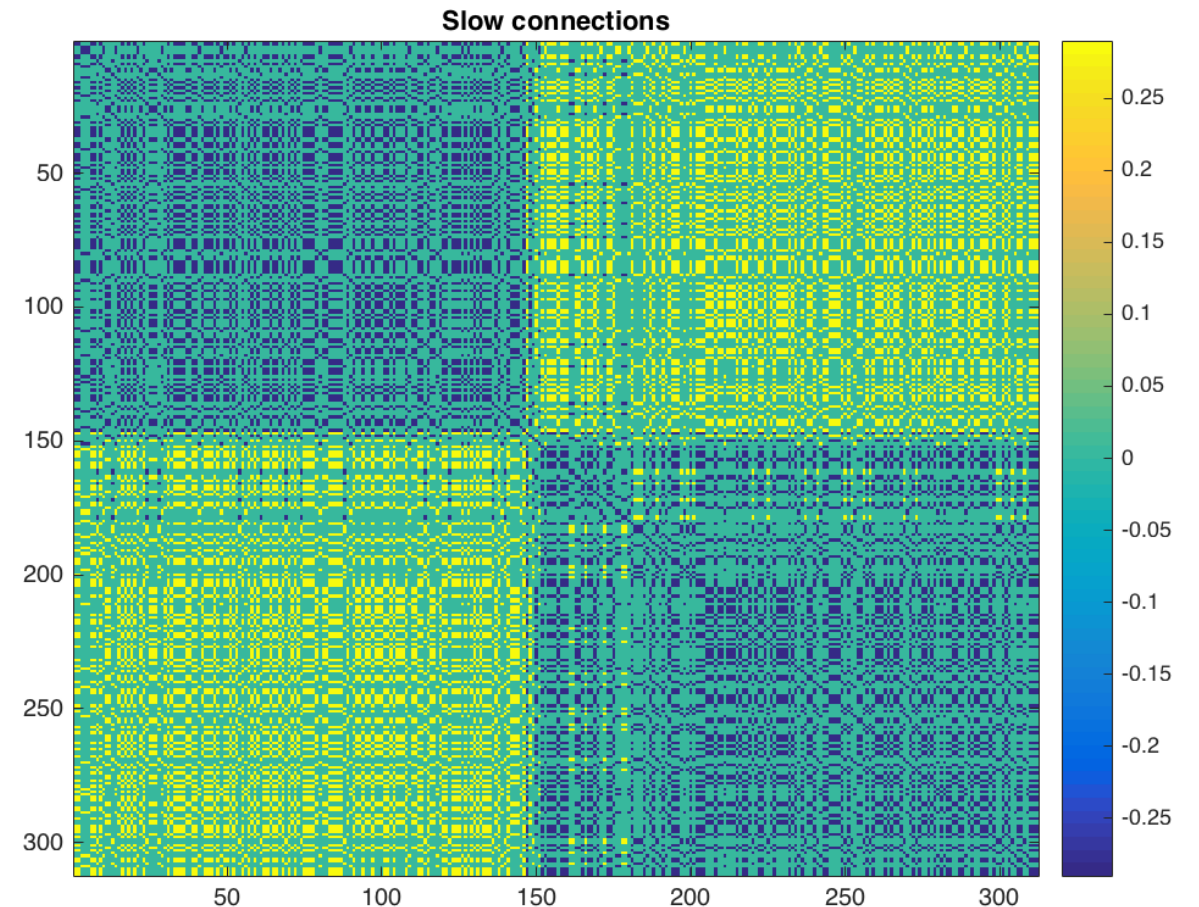
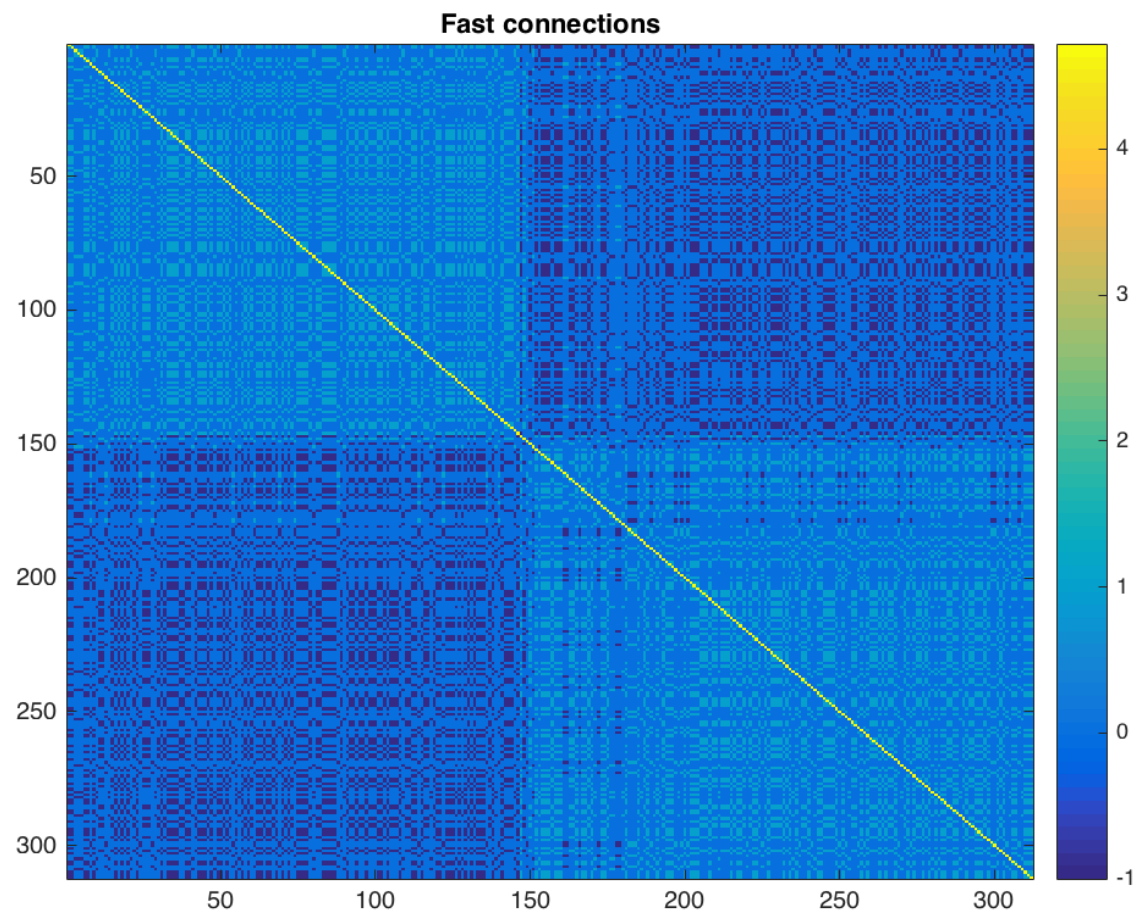
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PREDICTIVE CODING : SPIKING NETWORK

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No connectomics data to compare with but a way to compare with data would be to have a look at correlations matrix, chasing to average activity at a certain time precision.

DATA : RESPONSES VARIABILITY

Noise Correlations, Signal Correlations

$$\Theta = [\frac{\pi}{2}, 0, -\frac{\pi}{2}, -\pi]$$

$$\forall \theta \in \Theta$$

$$R_{\theta} = [R_{\theta}^1 \dots R_{\theta}^n] \in \mathbb{R}^n \text{ with } n = n_{\text{trials}}$$

computed for each neuron.

Noise correlation : average over the stimuli of the inter-trial correlation, computed stimulus by stimulus.

$$\forall (i, j) \in [1..N_{\text{neurons}}]^2$$

$$\rho_{i,j}^{\text{noise}} = \frac{\sum_{\theta \in \Theta} \text{corr}(R_{\theta i}, R_{\theta j})}{|\Theta|}$$

DATA : RESPONSES VARIABILITY

Noise Correlations, Signal Correlations

$$\Theta = [\frac{\pi}{2}, 0, -\frac{\pi}{2}, -\pi]$$

$$\forall \theta \in \Theta$$

$$\bar{R} = [\bar{R}_{\frac{\pi}{2}}, \bar{R}_0, \bar{R}_{-\frac{\pi}{2}}, \bar{R}_{-\pi}] \in \mathbb{R}^{N_{neurons} \times |\Theta|}$$

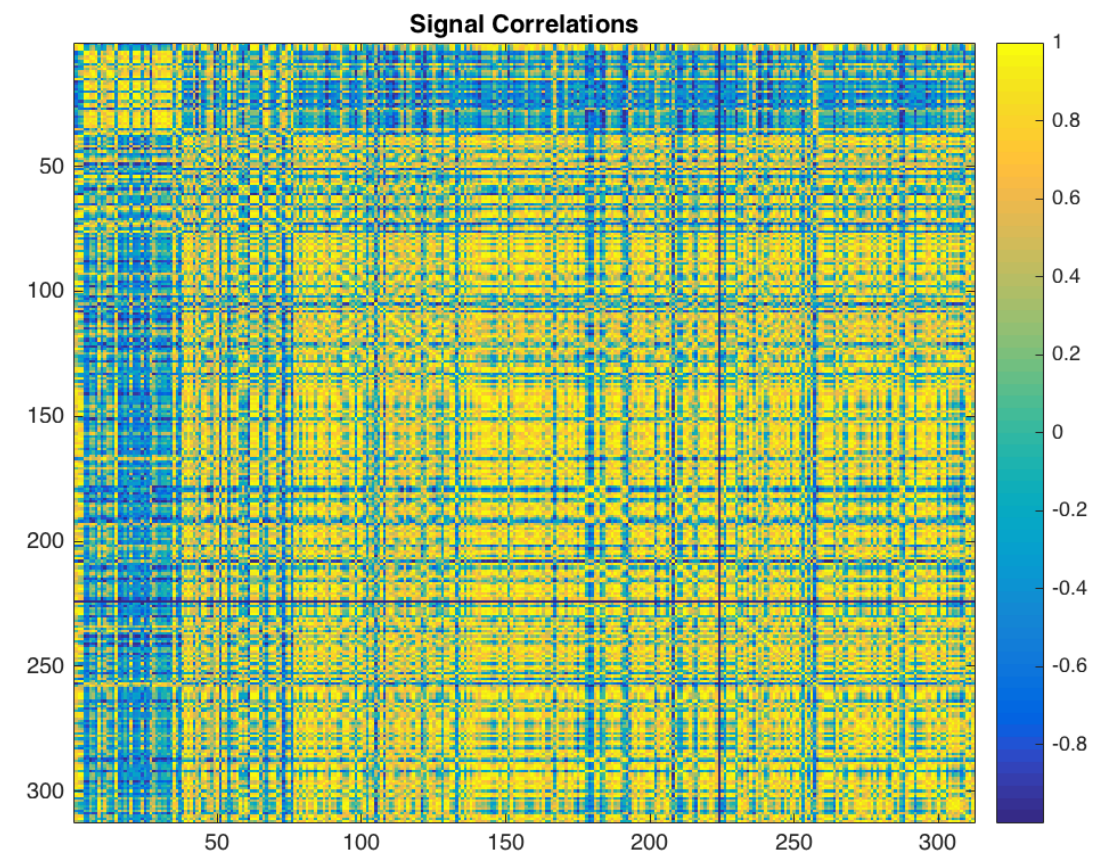
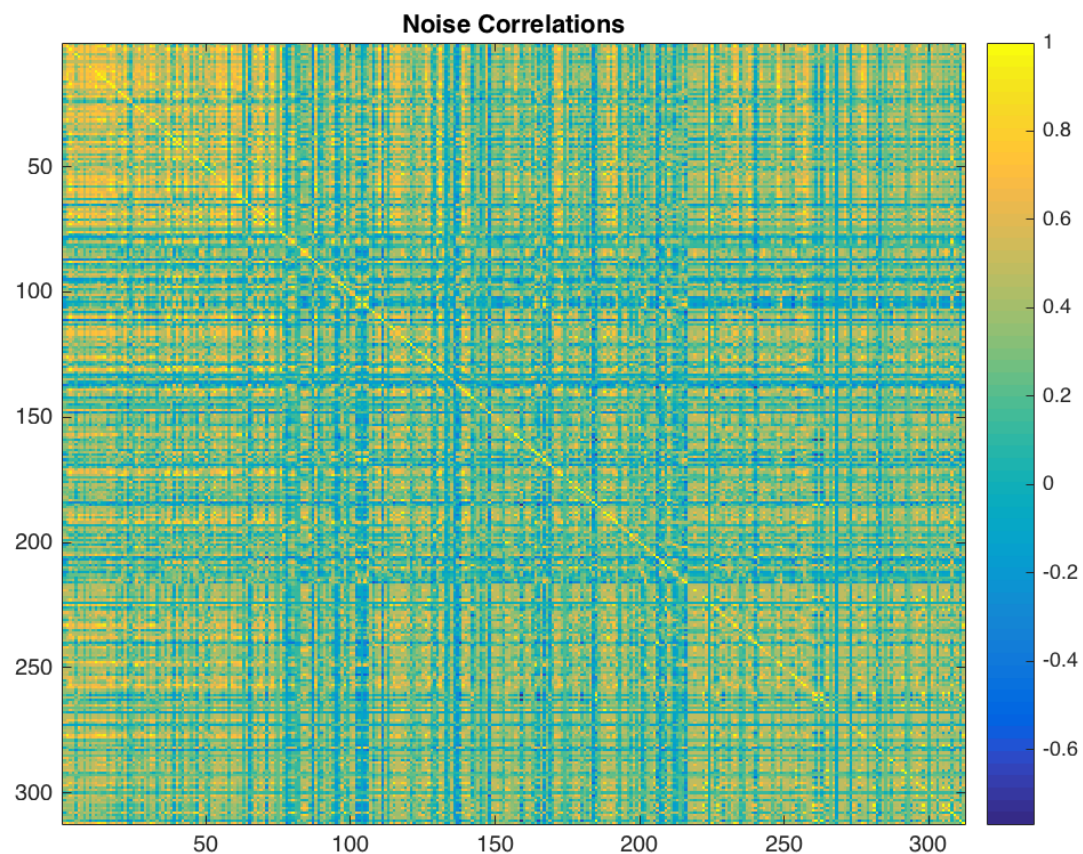
Signal Correlation : correlations between neurons average responses.

$$\forall (i, j) \in [1..N_{neurons}]^2$$

$$\rho_{i,j}^{signal} = \text{corr}(\bar{R}_i, \bar{R}_j)$$

DATA : RESPONSES VARIABILITY

Noise Correlations, Signal Correlations

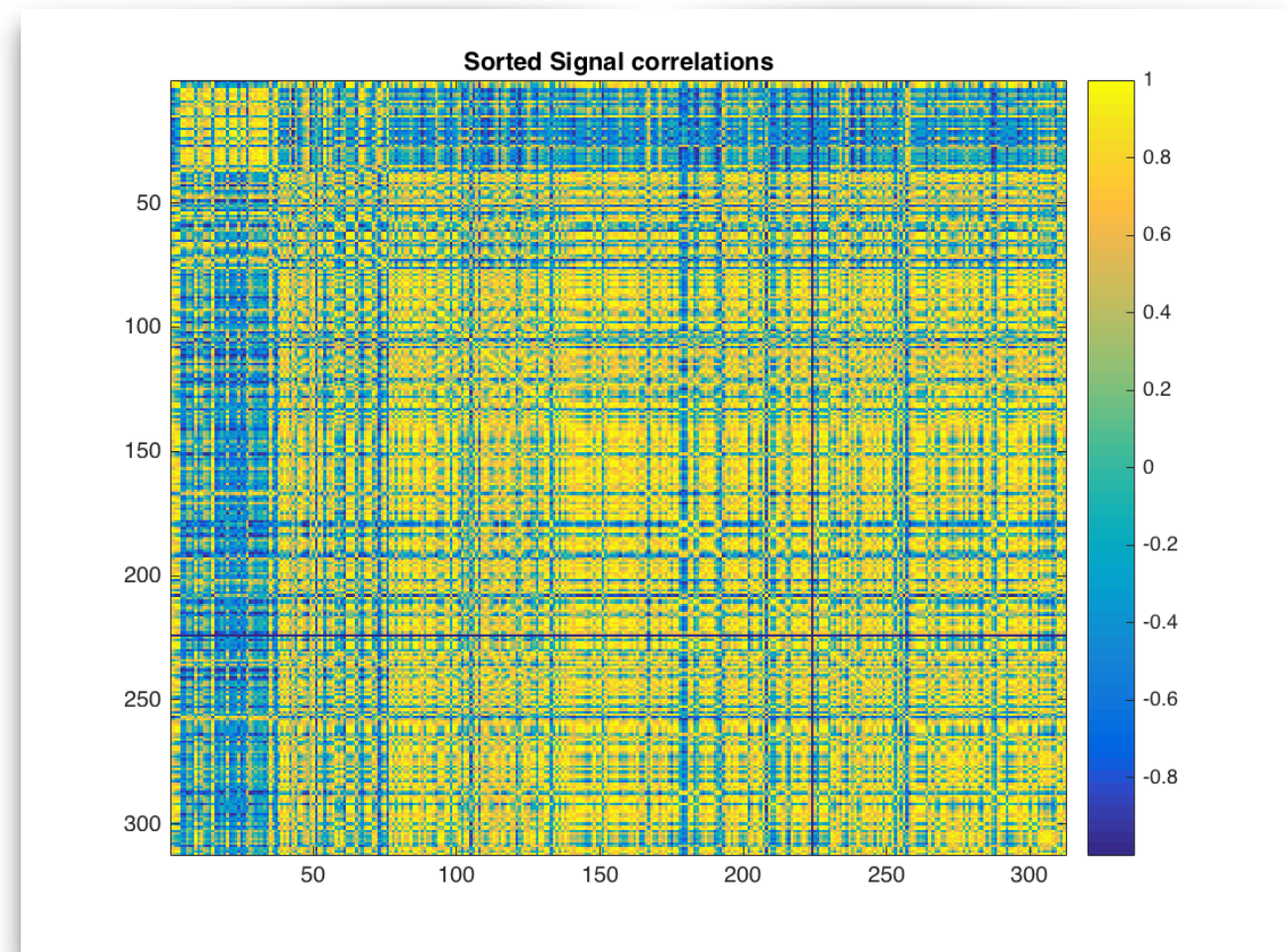


Noise correlation : average over the stimuli of the inter-trial correlation, computed stimulus by stimulus.

Signal Correlation : correlations between neurons average responses.

DATA : RESPONSES VARIABILITY

Noise Correlations, Signal Correlations



Noise correlation : average over the stimuli of the inter-trial correlation, computed stimulus by stimulus.

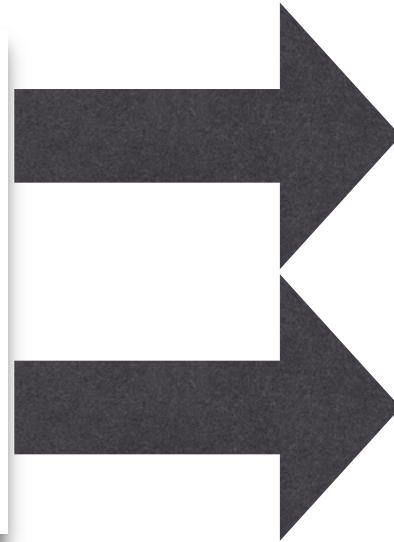
Signal Correlation : correlations between neurons average responses.

DATA : RESPONSES VARIABILITY

Noise Correlations, Signal Correlations

$$\mathbf{\Omega}^f = \mathbf{\Gamma}^T \mathbf{\Gamma} + \mu \lambda_d^2 \mathbf{I}$$

$$\mathbf{\Omega}^s = \mathbf{\Gamma}^T (\mathbf{A} + \lambda_d \mathbf{I}) \mathbf{\Gamma}$$



Analytical link with Noise
Correlation?

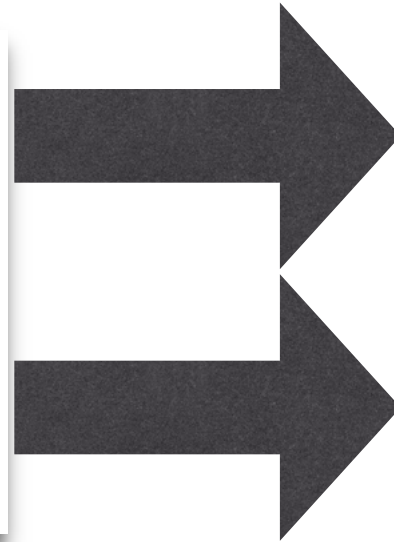
Analytical link with Signal
Correlation?

DATA : RESPONSES VARIABILITY

Noise Correlations, Signal Correlations

$$\mathbf{\Omega}^f = \mathbf{\Gamma}^T \mathbf{\Gamma} + \mu \lambda_d^2 \mathbf{I}$$

$$\mathbf{\Omega}^s = \mathbf{\Gamma}^T (\mathbf{A} + \lambda_d \mathbf{I}) \mathbf{\Gamma}$$



Analytical link with Noise Correlation?

Analytical link with Signal Correlation?

Are all correlations informative ?
Could we sort out between
informative correlation patterns
and non-informative ones ?

Consequence:

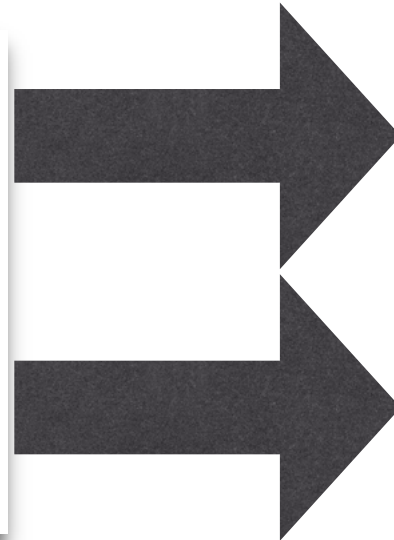
- determination of subnetworks
- relevancy for stimulus encoding
- size of the encoding network
kernels inference

DATA : RESPONSES VARIABILITY

Noise Correlations, Signal Correlations

$$\mathbf{\Omega}^f = \mathbf{\Gamma}^T \mathbf{\Gamma} + \mu \lambda_d^2 \mathbf{I}$$

$$\mathbf{\Omega}^s = \mathbf{\Gamma}^T (\mathbf{A} + \lambda_d \mathbf{I}) \mathbf{\Gamma}$$



Analytical link with Noise
Correlation?

Analytical link with Signal
Correlation?

Rush Die

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ACKNOWLEDGEMENTS

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SOME REFERENCES

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Information-limiting correlations. R.Moreno-Bote, J.Beck, I.Kanitscheider, x.Pitkow, P.Latham, A.Pouget.

